

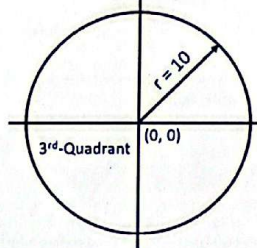
University of Chittagong
Department of Computer Science and Engineering
7th Semester B.Sc. (Engg) Examination-2023
Course Code: CSE-715 Course Title: Computer Graphics
Total marks: 54 Marks Time: 4.00 hours

[Answer any *three* questions from each of *Section-A* and *Section-B*. A separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks.]

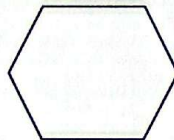
Section-A

1. Answer the following questions: 9 × 1 = 9
- How can you differentiate the computer graphics from image processing?
 - If we want to cut a 786×660 sub-image out from the center of a 1200×1200 image, what are the coordinates of the pixels in the large image that is at the lower left corner and upper right corner of the small image?
 - How does Bresenham's line algorithm overcome the limitations of the DDA algorithm?
 - Suppose, you are taking a picture of floating clouds from an open field. What kind of coordinate system is required to analyze the scenario?
 - State the advantages of matrix form for two or three-dimensional transformations.
 - You intend to draw a picture of your university, what kind of projection techniques you will consider?
 - Name the three perceptual terms for describing color and the corresponding physical properties of light.
 - Why do hidden-surface algorithms are needed?
 - Differentiate among the isometric, dimetric, and trimetric projections.

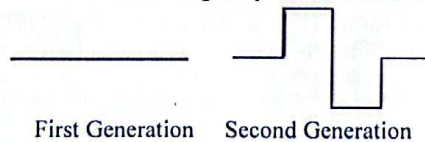
2. a) Why slanted lines look comparatively dimmer than straight lines? 1
- b) Indicate which raster locations would be chosen in the 3rd-Quadrant when scan converting the below circle using the Midpoint circle algorithm. 4



- c) Could you fill the following figure using flood fill algorithm with 8-connected pixels? Explain with the necessary diagram. 3



- d) A type of Koch curve can be drawn following the procedure below: 1



First Generation Second Generation

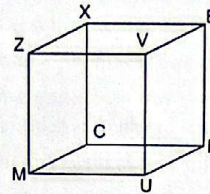
Now, draw the third generation of the curve.

3. a) While capturing and displaying images we need to keep in mind that monitor sizes may vary from one system to another. To resolve this issue, what kind of coordinate system is required and why? 1
- b) Find the new coordinates of the triangle P(-1, 2), Q(4,4), and R(1,-4) after 45° rotation for the following cases: (i) about the origin and (ii) about the point Q(4,4). 4
- c) Given a 3D object with coordinate points A(3, 0, 3), B(3, 3, 6), C(3, 0, 1), and D(0, 0, 0). Apply the scaling parameter 3 towards X axis, 2 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object. 4

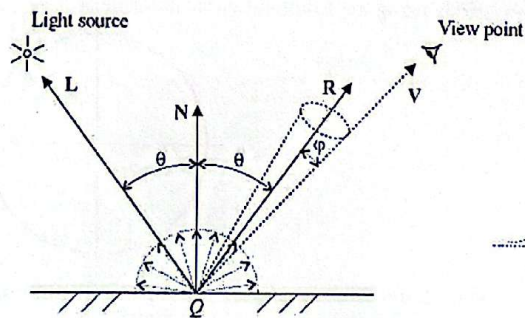
4. a) A polygon is defined by the following vertices in counterclockwise order: $A(1,1)$, $B(4,1)$, $C(4,4)$, and $D(1,4)$. The clipping window is defined by the rectangle with vertices: $W(2,2)$, $X(3,2)$, $Y(3,3)$, and $Z(2,3)$. Using the Sutherland-Hodgman algorithm, calculate the vertices of the clipped polygon step by step. Show your work clearly for each edge of the clipping window.
- b) Differentiate the orthographic and oblique projection. 1
- c) Explain the four anomalies of perspective projection with necessary diagrams. 4

Section-B

5. a) Find the transformation for (a) cavalier with $\theta = 60^\circ$ and (b) cabinet projections with $\theta = 45^\circ$. 3.5
- b) Represent the following cube by using polygon listing, explicit vertex listing, and explicit edge listing. 2.5



- c) Derive the equation of parallel projection onto the xy plane in the direction of projection $V = aI + bJ + cK$. 3
6. a) Given points $A(5, 1, -2)$, $B(10, 4, 9)$, and $C(15, -2, -3)$ and a viewpoint $V(0, 1, -10)$, determine which points obscure the others when viewed from V . 3
- b) Why is it easier to locate hidden surfaces when parallel projection is used? 1.5
- c) What is the difference between a vector and a ray? 1.5
- d) In light of the Phong model explain the following image. 3



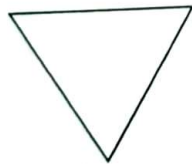
7. a) $C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$. Refer to this expression, what are the conditions whether a point is left or right of a line segment? 2
- b) Assume that a monitor produces the standard white D_{65} with $x_w = 0.3$, $y_w = 0.2$, and $Y_w = 1.0$ when $R = G = B = 1$, and the chromaticity coordinates of its phosphors are $R(0.62, 0.34)$, $G(0.29, 0.59)$, and $B(0.15, 0.06)$. Find the color transformation matrix M for the monitor. 3
- c) If point p_1 on line 3 has an RGB color $(5, 5, 0)$ and point p_2 on line 9 has an RGB color $(2, 0.5, 7)$, what is the color for line 5? 2
- d) Differentiate between the Painter's algorithm and the Z-buffer algorithm. 2
8. a) Perform the Cohen-Sutherland algorithm for a line segment with endpoints $(5,5)$ and $(15,10)$ in a clipping window with boundaries $(X_{min}=0, Y_{min}=0, X_{max}=10, Y_{max}=10)$. 3.5
- b) Derive the matrix for performing a perspective projection using homogeneous coordinates. 2
- c) Let, S_1 be a sphere of radius 8 centered at $(2, 4, 1)$ and S_2 a sphere of radius 10 centered at $(10, -2, -5)$. Determine if a ray with $s = 2I + 5K$ and $d = I - 2J$ intersects the spheres. 3.5

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[Answer any **three** questions from each of **Section-A** and **Section-B**. A separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks.]

Section-A

1. a) How do the following two fields distinguish from computer graphics? 2
(a) Image Processing (b) Computer-Human interaction.
- b) Why it is necessary to maintain a standard aspect ratio? If we want to resize a 1680×1050 image to one that is 1024 pixels wide with the same aspect ratio. What would be the height of the resized image? 2
- c) If we use 2-byte pixel values in a lookup table representation, how many bits does the lookup table occupy? 1
- d) If we use direct coding of CMY values with 3 bits for cyan and magenta and 4 bits for yellow, how many possible colors do we have for each pixel? 1
- e) Can a 5 by $3\frac{1}{2}$ inch image be presented at $3\frac{1}{2}$ by 4 inches without introducing geometric distortion? 2
- f) Why color printers are using subtractive color model? 1
2. a) What do you mean by the 8-way symmetry of a circle? Explain with diagram. 1.5
- b) Indicate which raster locations would be chosen by midpoint circle algorithm when scan converting a circle of radius 10 and center at (50, 50). 4
- c) Why slanted lines look comparatively dimmer than the straight lines? How do you solve this? 2
- d) A type of Koch curve can be drawn following the procedure below: 1.5



First Generation



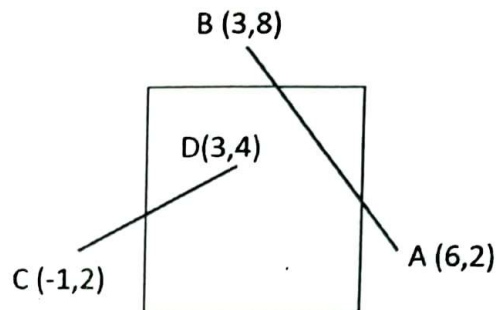
Second Generation

Now, draw the third generation of the curve.

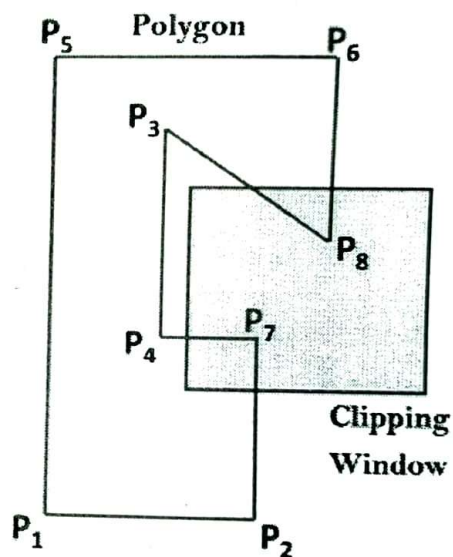
3. a) While capturing and displaying images we need to keep in mind that monitor sizes may vary from one system to another. To resolve this issue, what kind of coordinate system is required and why? 1
- b) Find the new coordinates of the triangle P(-1, 2), Q(2,4), and R(0,0) keeping Q(2,4) fixed for the following cases: (i) It has been expanded thrice its size and (ii) reduced to half its size. 4

- c) Find the matrix to perform a 45° rotation of the triangle A(5,6), B(2,1), and C(5,3) (i) about the origin and (ii) about a point (2, 3).

4. a) Find the region codes for end point A(6,2); B(3,8); C(-1,2) and D(2,4). Then, Clip the line segments AB and CD using Cohen-Sutherland algorithm where $X_{min}=2$, $X_{max}=5$, $Y_{min}=1$, and $Y_{max}=9$.



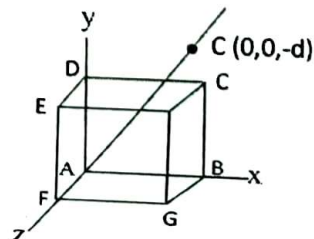
- b) Using the Sutherland-Hodgman polygon clipping algorithm, show the steps of clipping the following polygon.



- c) Find the normalization transformation from the window whose lower left corner is at (1,4) and upper right corner is at (3,4) onto the normalized device screen so that aspect ratios are preserved.

Section-B

5. a) Differentiate between perspective and parallel projection.
b) The unit cube in the following figure is projected onto the XY plane. Find the transformation and projection matrix for (a) cavalier with $\theta = 30^\circ$ and (b) cabinet projections with $\theta = 45^\circ$.



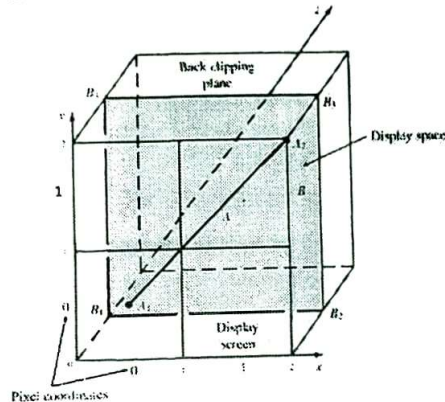
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(11) if $(t \leq 0)$ goto (15)
(12) $t_2 = 7 * b$

(16) $t_3 = r_1 + r_2$
 $r = 8 * t_3$

ere new basic blocks start. For each basic block
ion in the line is the first

- c) Define tilting as a rotation about the x-axis followed by a rotation about the y-axis. 3.5
 i) Find the tilting matrix. ii) Does the order of performing the rotation matter?
6. a) Given points A(1, 5, -2), B(1, 4, 9), and C(-2, -2, -3) and a viewpoint V(0, 1, -10), determine which points obscure the others when viewed from V. 3
 b) Why are hidden surface algorithms needed? 1.5
 c) What is the maximum number of objects that can be presented by using the Z-buffer algorithm? Using a 2x2 pixel display, show how the Z-buffer algorithm would determine the color of each pixel for the given objects A and B in the below figure. 4.5



7. a) $C = (x_2 - x_1)(y - y_1) - (y_2 - y_1)(x - x_1)$. Refer to this expression, what are the conditions whether a point is left or right of a line segment? 1
 b) Draw the example diagram of the convex and concave polygon. 1
 c) What is the purpose of homogeneous coordinates? How ray tracing technique incorporates the benefits of hidden surface removal and projection techniques? 3
 d) If point p1 on line 13 has an RGB color (0.4, 5, 0) and point p2 on line 15 has an RGB color (2, 0.5, 7), what is the color for line 12? 2.5
 e) Enumerate the significance of diffuse and specular reflection in the Phong illumination model. 1.5
8. a) Show that a ray is not a vector. 1
 b) Explain the color Gamut of the monitor. 1
 c) A ray is represented by $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ where $\mathbf{s} = 2\mathbf{i} - 3\mathbf{j} - 7\mathbf{k}$ and $\mathbf{d} = \mathbf{i} - \mathbf{j} - 2\mathbf{k}$. Find the coordinates of the points on the ray that corresponds to $t = 1, 2, -4, 3, 7$, and 5 , respectively. 3
 d) Assume a sphere of radius 2 with center located at the origin. If a ray originates from the point (3, 0, 0) in the direction (-3, 1, 0), determine the intersection point(s) of that ray with the sphere. 4

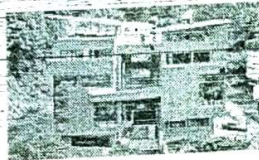
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[Answer any **three** questions from each of **Section-A** and **Section-B**. A separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks.]

Section-A

1. a) How does the image processing field differ from computer graphics? Briefly explain. 1
- b) If we use 2-byte pixel values in a 24-bit lookup table representation, how many bits does the lookup table occupy? 1.25
- c) If we use direct coding of RGB values with 12 bits per primary color, how many possible ways do we have for each pixel? 1
- d) You are asked to present the following image on two display devices whose sizes are given below. 3

1024×675



800×527

1800×1100

(a)

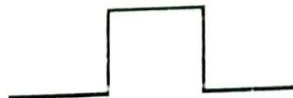
(b)

What kind of phenomena will be happened in each case? Explain with necessary computation.

- e) If we want to cut a 700×700 sub-image out from the center of a 900×800 image, what are the coordinates of the pixel in the large image that is at the upper right corner of the small image? 1.5
- f) A type of Quadratic Koch curve can be drawn following the procedure below: 1

The curve of generation 0 is a straight line.

The first generation of the curve can be drawn from the generation 0 curve by dividing it into three equal parts and replacing the middle part with a square-shaped bump as follows:



Now, draw the second generation of the curve.

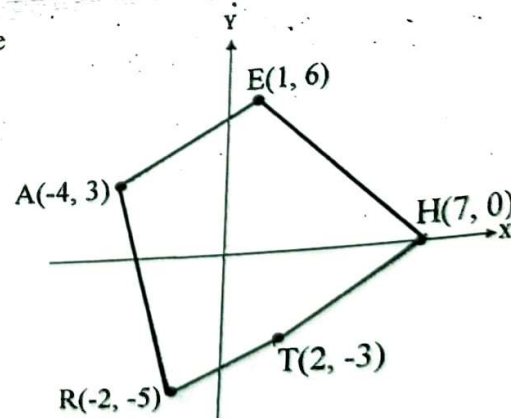
2. a) Suppose a straight line has endpoints (1, 1) and (10, 7). Now, answer the following questions. 1.5+1+1
+2.5=6
- Indicate which raster location would be chosen while scan converting this straight line using a basic line algorithm.
 - While doing (a), you find out that the scan conversion is taking place slower than you wanted. What might be the probable reason behind this?
 - Explain how can Bresenham line algorithm overcomes this limitation.
 - Indicate which raster location would be chosen while scan converting this straight line using the Bresenham line algorithm.
- b) What are the three major adverse side effects of scan conversion? In lowpass filtering, we reassign the pixel values using a filter/kernel. Considering this, reassign the values of the below pixel grid using the given kernel. 2.75

0	0	0	0	0
0	13	31	46	0
0	25	92	31	0
0	44	74	94	0
0	0	0	0	0

0	3	0
0	0	0
0	2	0

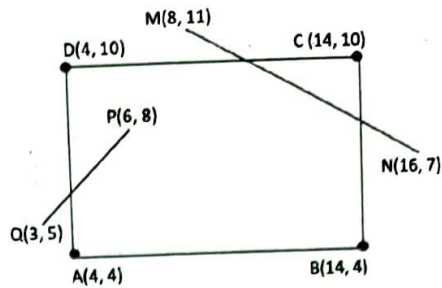
Kernel

3. a) Differentiate between the geometric and coordinate transformation. 1.25
- b) Magnify the Pentagon ARTHE to twice its size while keeping E(1, 6) fixed. 3.5

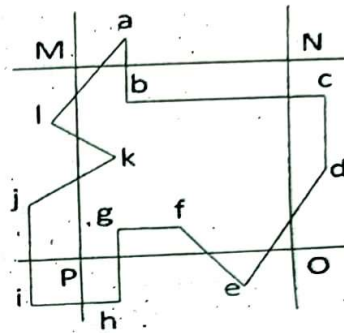


- c) Perform a 60° rotation of the rectangle A(1, 1), B(1, 2), C(2, 2), and D(2, 1) about the point E(-1, -1). 4

4. a) Find the region codes for endpoints M, N, P, and Q. Then, Clip the line segments MN and PQ using the Cohen-Sutherland algorithm.



- b) Let abcdefghijkl and MNOP be the polygon to be clipped and the viewing window respectively as shown in the given figure. Using the Sutherland-Hodgman polygon clipping algorithm, show the steps of clipping the polygon in the following order of clipping edges:



- a) Left clip edge PM
b) Bottom clip edge PO
c) Identify the convex and concave polygons. Briefly explain your selection.

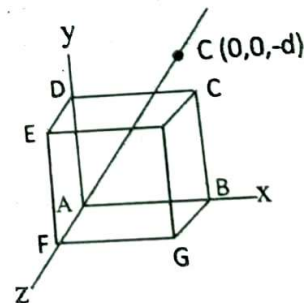


Section-B

5. a) The Cartesian coordinate of a pixel is $P(1,2)$. Identify the valid coordinates in homogeneous space from the following options.

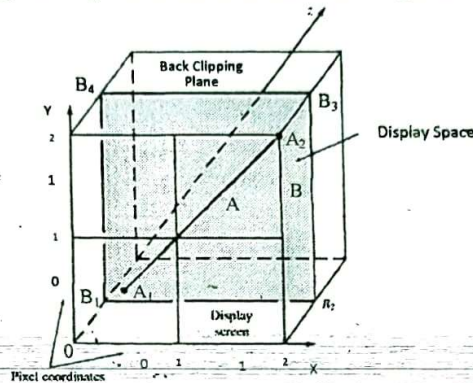
- i) $P(1, 2, 0)$ ii) $P(1, 2, 1)$ iii) $P(4, 2, 4)$ iv) $P(4, 2, 1)$

- b) The unit cube in the following figure is projected onto the XY plane. Find the transformation and projection matrix for (a) cavalier with $\theta = 45^\circ$ and (b) cabinet projections with $\theta = 30^\circ$.

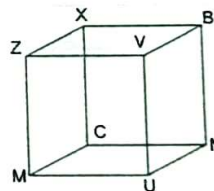


- c) What do you mean by perspective foreshortening and vanishing point? Explain with the necessary diagram.
d) What is the difference between a local illumination model and a global illumination model?

6. a) Given points $P_1(1, 3, 1)$, $P_2(3, 6, -11)$, and $P_3(2, 6, -5)$ and a viewpoint $C(0, 0, 7)$, determine which points obscure the others when viewed from C . 3
- b) Distinguish between perspective and parallel projections. Which projection is used by architects and engineers. 1
- d) What is the maximum number of objects that can be presented by using the Z-buffer algorithm? Using a 2×2 pixel display, show how the Z-buffer algorithm would determine the color of each pixel for the given objects A and B in the below figure. 4.75



7. a) Assume that a monitor produces the standard white D_{65} with $x_w = 0.313$, $y_w = 0.329$, and $Y_w = 1.0$ when $R = G = B = 1$, and the chromaticity coordinates of its phosphors are $R(0.62, 0.34)$, $G(0.29, 0.59)$, and $B(0.15, 0.06)$. Find the color transformation matrix M for the monitor. 3
- b) Represent the following cube by using polygon listing and explicit vertex listing. 1.75



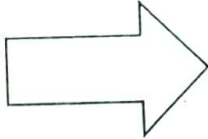
- c) If point P_1 on line 6 has an RGB color $(1, 1, 0)$ and point P_2 on line 15 has an RGB color $(0.6, 0.5, 0.2)$, what is the color for line 8? 2
- d) Find the normalization transformation that maps a window whose lower left corner is at $(1, 1)$ and upper right corner is at $(2, 2)$ onto a viewport that has a lower left corner at $(0, 0)$ and upper right corner $(1/2, 1/2)$. 2
8. a) Describe how hidden surface removal and projection are integrated into the ray-tracing process. 1.5
- b) A ray is represented by $\mathbf{r}(t) = \mathbf{s} + t\mathbf{d}$ where $\mathbf{s} = 2\mathbf{I} + \mathbf{J} - 3\mathbf{K}$ and $\mathbf{d} = \mathbf{I} + 2\mathbf{K}$. Find the coordinates of the points on the ray that corresponds to $t = 0, 1, 7, 4$, and 3 , respectively. 2.5
- c) What is the difference between a vector and a ray? 1
- e) Let S_1 be a sphere of radius 8 centered at $(2, 4, 1)$ and S_2 a sphere of radius 10 centered at $(10, -2, -5)$. Determine if a ray with $\mathbf{s} = 2\mathbf{J} + 5\mathbf{K}$ and $\mathbf{d} = \mathbf{I} - 2\mathbf{K}$ intersects the spheres. 3.75

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[Answer any **three** questions from each of the **Group-A** and **Group-B**. A separate answer script must be used for Group-A and Group-B. Figures in the right-hand margin indicate full marks.]

Group-A

1. a) Which discipline describes producing and synthesizing digital images? Justify your answer. 3
 - b) Can a 5 by 3½ inch image be presented at 3½ by 4 inches without introducing geometric distortion? 2
 - c) How a digital image can be formed from object definition? 1.25
 - d) Why it is necessary to maintain a standard aspect ratio? If an image has a height of 2 inches and an aspect ratio of 1.5, what is its width? 1+1.5

 2. a) What do you mean by subtractive color model? Give an example. Why do many color printers use a separate black ink cartridge? 1+1
 - b) If we use direct coding of RGB values with 10 bits per primary color, how many possible colors do we have for each pixel? 1.5
 - c) If we want to cut a 700×700 sub-image out from the center of a 900×800 image, what are the coordinates of the pixel in the large image that is at the upper-right corner of the small image? 2.75
 - d) Could you fill the following figure using the Flood Fill algorithm with 8-connected pixels? Justify your answer. 2.5
- 
3. a) Find the CMY coordinates of a color at (0.1, 0, and 0.5) in the RGB space. 1
 - b) Show raster locations while drawing a line using the DDA algorithm from (1, 1) to (5, 8). 3.75
 - c) Find raster locations using Bresenham's algorithm when scan-converting a line from pixel coordinate (0, 0) to a pixel coordinate (8, 5). 4

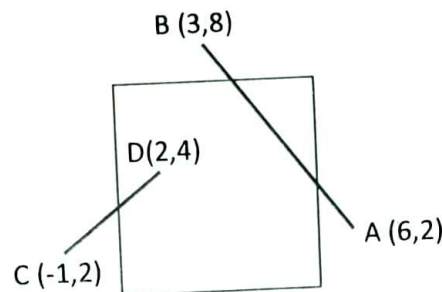
 4. a) Perform a 90° rotation of rectangle A (1, 1), B (1, 2), C (2, 2), and D (2, 1) about the point B (-1, -1). 4.25
 - b) Magnify the following triangle with vertices A (1,1), B (2, 1), and C (2, 2) to twice its size while keeping B (2, 1) fixed. 3.5
 - c) Show 8-way symmetry of a circle. 1

Group-B

5. a) Find the workstation transformation that maps the normalized device screen onto a physical device whose x extent is 0 to 199 and y extent is 0 to 639 where the origin is located at the lower-left corner. ??? 2
- b) Find the normalization transformation that maps a window whose lower-left corner is at (1, 1) and upper right corner is at (2, 2) onto a viewport that has a lower-left corner at (0, 0) and upper right corner (1/2, 1/2). 2.75
- c) Briefly explain the Cohen Sutherland algorithm. 3
- d) Differentiate between the geometric and coordinate transformation. 1

6. a) While capturing and displaying images we need to keep in mind that monitor sizes may vary from one system to another. To resolve this issue, what kind of coordinate system is required and why? 1

b)

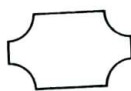


Find the region codes for endpoint A(6,2), B(3,8), C(-1,2), and D(2,4). Clip the line segments AB and CD using the Cohen-Sutherland algorithm. (Assume $X_{min}=1$, $X_{max}=5$, $Y_{min}=1$, $Y_{max}=7$)

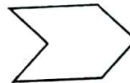
- c) Identify the convex and concave polygons from the below figures and justify your selection. 2



(i)



(ii)



(iii)



(iv)

- d) What do you mean by perspective foreshortening and vanishing point? 1.75

7. a) Show the conditions to identify whether a point locates the right or left side of a line segment. 2.25
- b) Define positively and negatively oriented convex polygon. 3.5
- c) Demonstrate 3D rotation matrices. ??? 3

8. a) Distinguish between perspective and parallel projections? Which projection is used by the architects and engineers? 3
- b) Draw the diagrams of isometric, diametric, and trimetric projections. 2.25
- c) Given points $P_1(1, 2, 0)$, $P_2(3, 6, 20)$, and $P_3(2, 4, 10)$ and a viewpoint $C(0, 0, -10)$, determine which points obscure the others when viewed from C. 3.5